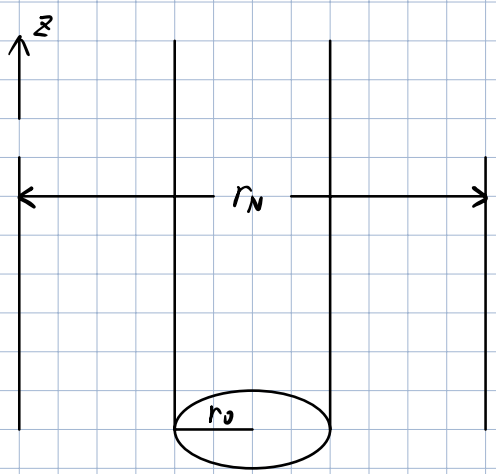
$$N(z) = N_k \cos(kz)$$

$$\nabla^2 \varphi = 0$$

$$\nabla^2 = \frac{\partial}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}$$

$$\phi = R(r)\, e^{ikz}.$$

$$R'' + \frac{1}{r}R' - k^2 R = 0.$$



$$R(r) = A_k K_0(kr) + B_k I_0(kr)$$

② 식 정리 : $\nabla^2 \varphi = \dfrac{N_k g \cos kz}{\varepsilon_0 \pi r_0^2}$

$$R(r) = C_k K_0(kr) + D_k I_0(kr) + \frac{N_k g \cos kz}{\varepsilon_0 \pi r^2}$$

$R(r = r_N) = 0.$

当 $r = r_0$，由连续性有：

$$R(r_0) = A_k K_0(kr_0) + B_k I_0(kr_0) = C_k K_0(kr_0) + D_k I_0(kr_0) + \frac{N_k g \cos kz}{\varepsilon_0 \pi r_0^2}$$

$$R'(r_0) = R'(r_0).$$

$$\begin{cases} D_k I_0(kr_0) + \dfrac{gN_k}{\pi r_0^2 \varepsilon_0 k^2} = A_k \chi_0(k), & \chi_0 \equiv k_0(kr_0) - \dfrac{k_0(kr_0)}{I(kr_0)} I(kr_0) \\ -D_k I_0(kr_0) = A_k \chi_1(k), & \chi_1 \equiv k_1(kr_0) + I_1(kk_0) \end{cases}$$

解得：

$$\begin{cases} D_k = -\dfrac{\chi_1(k)}{I_1(kr_0)} A_k \\ A_k = \dfrac{gN_k}{\pi r_0^2 \varepsilon_0 k^2} \dfrac{1}{\chi_0(k) + \dfrac{I_0(kr_0)}{I_1(kr_0)} \chi_1(k)} \end{cases}$$

$$\int_{-r_0-\delta R}^{r_0+\delta R} E_r\, dr = -[\phi_{k,1}(r_0+\delta R, z) - \phi_{k,1}(r-\delta R, z)]$$

$$= -[\phi_{k,2}(r_0+\delta R, z) - \phi_{k,2}(r-\delta R, z)].$$

$$E_{rsc} = -\frac{\phi_{k,2}(r+\delta R, z) - \phi_{k2}(r-\delta R, z)}{2r_0.} = -\frac{\delta R}{r_0}\frac{\partial \phi_{r,z}(r_0, z)}{\partial r}$$

$$= -\frac{\delta R}{r_0} k D_k I(kr_0)\cos(kz) = \alpha\, \delta R\, \omega_c^2 \frac{m}{q}$$

$$\alpha = \frac{-qkD_k I_1(kr_0)}{r_0 \omega m}\cos kz. = \frac{q^2 N(z)}{\pi \varepsilon_0 k r_0^3 \omega_r^2 m}\frac{\chi_1(k)}{\chi_0(k) - 1\frac{I_0(kr)}{I_1(kr)}\chi_1(k)}$$

$$\gamma_t^2 = \frac{\frac{\delta P}{P}}{\frac{\delta R}{R}}$$

$$q(v+\delta v)B\left(1-n\frac{\delta R}{R}\right) = \frac{(\gamma+\delta\gamma)m_0(v+\delta v)^2}{R+\delta R} + qE_{rsc}$$

$$\frac{1}{R}\frac{1}{1+\frac{\delta R}{R}} = \frac{1}{R}\left(1-\frac{\delta R}{R}\right).$$

$$\therefore -qvBn\frac{\delta R}{R} + qB\delta v = -\gamma m_0 v^2 \frac{1}{R}\frac{\delta R}{R} + m_0 v^2 \frac{1}{R}\delta r + \gamma m_0 2v\frac{1}{R}\delta v.$$
$$+ \alpha m \omega_c^2 \delta R$$

$$\delta P = (r+\delta r)m_e(v+\delta v) - r m_0 v = m_0 v\,\delta r + m_0 \gamma\,\delta v + m_0\,\delta\gamma\,\delta v$$

$$\gamma_t^2 = \frac{\frac{\delta P}{P}}{\frac{\delta R}{R}} = \frac{\frac{\delta\gamma}{\gamma} + \frac{\delta v}{v}}{\frac{\delta R}{R}} \qquad \frac{\delta\gamma}{\gamma} = \frac{\delta R}{R}\gamma_t^2 - \frac{\delta v}{v}$$

$$-qBn\delta R + qB R\delta v = -\frac{\gamma m_0 v^2}{R}\delta R + m_0 v^2\left(\frac{\gamma}{R}\gamma_t^2\delta R - \frac{\gamma}{v}\delta v\right).$$

$$+ 2\gamma m_0 v\delta v + \alpha m\omega_c^2 R\delta R.$$

$$\delta v(qBR - \gamma m_0 v) = \delta R\left(qvBn - \frac{\gamma m_0 v^2}{R} + \frac{m_0\gamma v^2}{R}\gamma_t^2 + \alpha m\omega_c^2 R\right).$$

$$\begin{cases} qBR = \gamma m_0 v \\ qvBn - \dfrac{\gamma m_0 v^2}{R} + \dfrac{\gamma m_0 v^2}{R}\gamma_t^2 + \alpha m\omega_c^2 R = 0 \end{cases}$$

$$\Rightarrow \qquad \gamma_t^2 = 1 - n - \alpha.$$